

RABIA SAMAD AFRIDI

AI Engineer — Conversational AI & Voice Agents | LLM Systems | Technical Delivery Lead

Dubai, UAE · +971 50 811 8657 · rabiasamad895@gmail.com

[[LinkedIn](#)] [linkedin.com/in/rabia-samad-ai/](https://www.linkedin.com/in/rabia-samad-ai/) · [[portfolio](#)] <https://rabiasamad.github.io/>

Nationality: Pakistani · **Visa:** UAE Family Sponsored Residence(work permit only) · Notice period: 30 days

PROFESSIONAL SUMMARY

AI Engineer with 3+ years shipping production LLM systems in healthcare and media, specializing in **real-time voice agents**, RAG architectures, and agentic automation. Delivered an end-to-end voice-AI platform for dental clinics handling high-volume patient scheduling, and a content-generation system that cut news production time by 80%. Proven technical delivery lead managing cross-functional teams of 9–15. MSc Electrical Engineering (AI & Autonomous Systems), NUST. Dubai-based since Feb 2026; available for immediate work-permit on boarding.

TECHNICAL SKILLS

AI / ML:

LLMs · Agentic AI · Retrieval-Augmented Generation (RAG) · Prompt Engineering · Fine-tuning · Text To Speech · Generative AI · Conversational AI · Voice Agent Design

Frameworks & Libraries:

LangChain · LangGraph · CrewAI · TensorFlow · PyTorch · FastAPI

Automation & Integrations:

ChatGpt · Claude · N8N · Zapier · ElevenLabs · Vapi · Retell · Twilio · HeyGen · Google AI Platform · whatsapp Business API · Instagram Dms Automations

Languages & Infrastructures:

Python · PostgreSQL · Vector DBs (Chroma, FAISS) · AWS EC2/S3 · Docker · CI/CD GitHub Actions

WORK EXPERIENCE

Technical Project Lead · [Smart Forum](#) (Pakistan Remote)

May 2025 – Present

Project: Apointlo — AI Voice Agent for Dental Clinics

- Architected production voice-AI system using VAPI, Twilio, SIP trunks, and N8N, handling 500+ patient calls/month across 2 clinics with automated scheduling, reducing receptionist workload.
- Integrated WhatsApp Business API for real-time appointment confirmations and reminders, improving patient show-up rates.
- Designed core voice call flows, and defined product requirements and user stories for the real-time monitoring dashboard delivered by the design/front-end team.
- Managed project delivery, timelines, and execution across a cross-functional team of 9 developers, designers, QA engineers, and DevOps specialists.

Project: Curio — Mobile App for Autism Early Detection

- Led an agile team of 15 (Developers, Designers, QA, DevOps, Domain Experts) from concept through launch; 1,500+ Android and 200+ iOS downloads.
- Built RAG-based chatbot for personalized guidance and support for parents (Multi-Lingual with Voice Input support).

- Built AI-powered therapy modules and interactive assessment tools for autism early detection.
- Collaborated with psychologists and therapists to ensure clinical accuracy and usability.
- Awarded Star Performer Award (Q4 2025) for outstanding delivery on a high-impact healthcare product.

AI Engineer · Smart Forum

August 2023 – May 2025

- Enhanced a lip-sync DL model through data collection, annotation, and preprocessing, resulting in a 60% improvement in output quality and a 50% reduction in inference latency.
- Developed an AI-powered content generation platform leveraging prompt engineering to produce news articles and scripts in a consistent broadcast-quality style.
- Designed and implemented automated publishing workflows using N8N enabling one-click content distribution to client websites.
- The platform is live and has cut news production time by 80%.

RESEARCH

Liver Cancer Detection Using Neural Networks

- Developed a neural-network-based system for early-stage liver cancer detection using ultrasound data.

Hand Detection from Images

- Applied classical computer vision techniques, including Hough Transform, Harris Corner Detection, and SIFT feature extraction, to identify and localize hands in images.

CERTIFICATIONS

- Generative AI with Large Language Models — DeepLearning.AI
- Prompt Engineering with Llama 2 & 3 — DeepLearning.AI
- Introduction to AI & Machine Learning on Google Cloud — Google Cloud Training
- AI Upskilling Certificate: Technical Foundation — Qualcomm
- AI Upskilling Certificate: Hands on development from model to App — Qualcomm
- LangChain for LLM Application Development — DeepLearning.AI
- Multi AI Agent Systems with CrewAI — CrewAI

EDUCATION

Master of Science — Electrical Engineering

Sep 2021 – Jan 2024

National University of Sciences & Technology (NUST), Pakistan

Specialization: Artificial Intelligence and Autonomous Systems

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Mobile Robotics, Stochastic Systems

Bachelor of Science — Electrical Engineering

Aug 2015 – Jun 2019

National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan

Specialization: Computer Engineering

Relevant coursework: Embedded Systems, Digital Signal Processing, Data Communications, Robotics, Computer Architecture

AWARDS

- Star Performer Award, Q4 2025 — Smart Forum
- Dean's List of Honor — Spring 2018, Fall 2018 and Spring 2019
- Bronze Medal for Academic Excellence — Fall 2017